

Table Schema and Description

1. Search Event Fact Table (search_event_fact)

Column Name	Data Type	Description
search_event_id	STRING	PK - Unique identifier for each search event.
search_event_timestamp	TIMESTAMP	Date and time when the search event occurred.
session_duration	FLOAT	Duration (in seconds) of the session.
has_ct	BOOLEAN	Indicates whether a click-through (CT) event occurred.
has_atc	BOOLEAN	Indicates whether an add-to-cart (ATC) event occurred.
has_purchase	BOOLEAN	Indicates whether a purchase event occurred.
search_results_count	INTEGER	Number of products displayed in the search event.
search_terms	STRING	The search keywords used by the user.
search_terms_type	STRING	Type of search (manual-search, type-ahead).
environment_type	STRING	Platform used (web-prod, ios-prod, android-prod).
search_models	STRING	Search ranking model (Model_A, Model_B).
device_class	STRING	Type of device used (mobile, laptop, tablet).
traffic_source	STRING	Source of traffic (Organic Search, Referral, Direct).
user_id	STRING	FK - linking to user_dim. Identifies the user performing the search.
store_id	STRING	FK - linking to store_dim. Identifies the store associated with the event.
event_tracking	ARRAY<STRUCT>	Nested structure containing ct_event_id, atc_event_id, and purchase_event_id.
ct_event_id	STRING	Unique ID for a click-through event (only present if has_ct = True).
atc_event_id	STRING	Unique ID for a click-through event (only present if has_atc = True).
purchase_event_id	STRING	Unique ID for a click-through event (only present if has_purchase = True).
product_details	ARRAY<STRUCT>	Nested structure containing product_id, fulfilment_type, cart_id, product_position, and num_purchase.
product_id	INTEGER	FK - Unique identifier for each product.
fulfilment_type	STRING	Method of order fulfillment chose by user (delivery, pickup, or courier).

num_purchase	FLOAT	Number of units/pounds purchased (only applies when purchase_event_id present).
product_position	INTEGER	The product's position in search results.

2. Product Dimension Table (product_dim)

Column Name	Data Type	Description
product_id	STRING	PK - Unique identifier for each product. Primary Key.
product_category	STRING	The category to which the product belongs (e.g., electronics, fashion).
product_name	STRING	The official name of the product.
product_price	FLOAT	The price of the product in USD.
product_sponsor	BOOLEAN	Indicates if the product is a sponsored listing.
product_rating	STRING	The brand name of the product.
product_sku	STRING	Unique identifier for each product.

3. Store Dimension Table (store_dim)

Column Name	Data Type	Description
store_id	STRING	PK - Unique identifier for each store. Primary Key.
store_city	STRING	The city where the store is located.
store_country	STRING	The country where the store is located.
store_manager_id	INTEGER	Unique 12-digit ID for the store manager.
store_rating	FLOAT	Rating of the store (1 to 5 scale).
store_status	STRING	Status of the store.

4. User Dimension Table (user_dim)

Column Name	Data Type	Description
user_id	STRING	PK - Unique identifier for each user. Primary Key.
membership_status	STRING	Type of user membership (Base, Premium, None.).
user_country	STRING	Country where the user is located (USA, Canada, Other).
user_age	INTEGER	The age of the user.
user_gender	STRING	The gender of the user (male, female).
user_language	STRING	Preferred language of the user (en, fr, es).